Homework 1

BIOSTAT 740, Fall 2026

Due September 26, 8:00 PM

Purpose:

1. Fit a generalized least squares model in SAS and R
2. Try to figure out what the output is showing you.
3. Report out some key features.

HW2 tasks

1. Use the tlc data set from Canvas that you downloaded last week.
2. Select (subset to) just the placebo group (trt = P) for parts 3-5 of this assignment.
3. Fit the (incorrect) general linear model [using SAS `proc mixed` or `proc glm` and R `lm()` function] and assess H_0: equal mean at each time point. Report the F statistic, degrees of freedom, p-value, and decision.
4. Fit the general**ized** lineal model to the data [using SAS `proc mixed` and R `gls()` function from the nlme package] and assess H_0: equal mean at each time point. Report the F statistic, degrees of freedom, p-value, and decision. Use the `summary()` function to see what the nlme package gives you as the output from the model.
5. SAS will (almost) always spit out huge amounts of data. The `gls()` function from nlme is similar. For this part of the assignment, write down one question about what a piece of the output is about, what is for, or what it means, and we will talk about it in class. You don't need to figure it out for yourself unless you want to.
6. Now, revert to the whole dataset, including both placebo (trt = P) and active (succimer, trt = A) groups.
7. In SAS `proc mixed` and r nlme `gls()` fit the model with both groups and follow the process outlined in class on Wednesday. What do you conclude about the two groups?

SAS hints

1. You will need `proc transpose` again to format the data as needed.
2. code for general linear model

```
proc mixed data = tlclong2;
class id time;
model y = time / s;
run;
```

3. code for generalized linear model

```
proc mixed data = tlclong2;
class id time;
model y = time / s;
repeated / type=un subject=id R Rcorr;
run;
```

R hints

1. Use `pivot_longer()` or `gather()` functions from the tidyverse package to convert the data to wide format. `pivot_longer(0)` is slightly less awkward, I think:

```
tlcplong <- filter(tlc,trt=="P") %>%
  pivot_longer(starts_with("y"), names_to = "time", values_to= "y")
```

2. Code for general linear model

```
tlc.glm = lm(y ~ relevel(as.factor(timev),4), data=tlcplong)
```

3. Code for generalized linear model

```
tlc.gls = gls(y ~ relevel(as.factor(timev),4), data=tlcplong,
   correlation = corSymm(form = ~ 1|id),
   weights = varIdent(form = ~1|timev))
```

What to turn in

Please submit **on Canvas** a single PDF file with

1. Annotated R code. (Annotate to explain what your code is doing.)
2. Annotated SAS code.
3. Your conclusion from the (incorrect) general linear model from part 3.
4. Your conclusion from the (correct) generalized linear model from part 4.
5. A question about the output from `proc mixed` or `gls()`.
6. Your conclusions about the differences between the two groups from part 7.

Cutting and pasting is fine! I encourage you to take screenshots of your SAS code and output to put into your .Rmd file.